N. Rejwana

Analysis of spatial transcriptomic data in human breast cancer

downloaded file from-
https://www.10xgenomics.com/datasets/fresh-frozen-visium-on-cytassist-human-breast-cancer-probe-based-whole-transcriptome-profiling-2-standard

10xgenomics.com/datasets/fresh-frozen-visium-on-cytassist-human-breast-cancer-probe-based-whole-transcriptome-profiling-2-standard

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Output files	File type	Size	md5sum
Genome-aligned BAM	BAM	3.72 GB	8927a54485cca0e4eded926df9e3dfd0
Genome-aligned BAM index	BAI	2.19 MB	1293220d825b6fc317cd68a488ed578a
Per-molecule read information	H5	253 MB	e2f2119b72aca64ed4329946551643ca
Feature / barcode matrix HDF5 (filtered)	H5	11.4 MB	5c6f495d69a566185ccf5a1e5edf76fb
Feature / barcode matrix (filtered)	GZ	34.5 MB	c94e614563ebeecba655158c32405da2
Feature / barcode matrix HDF5 (raw)	H5	16.6 MB	2f52bf85875fbf01506026310a84fb5e
Feature / barcode matrix (raw)	GZ	43.8 MB	4e37710326576f822a3990cba54b5f2a
Clustering analysis	GZ	24.6 MB	8c03133d9d68e581a029d00ba63c2cb5
Spatial imaging data	GZ	29.1 MB	68239b455af01fb99afacab814b432bc
Summary CSV	CSV	845 B	e9a56e984751a9163018f2111c4e31bd
Summary HTML	HTML	5.02 MB	61453ed6fa53fd4e110ae2efb4e07247
Loupe Browser file	CLOUPE	1.11 GB	9aa1d8763fd1851bb7a2924276d94aa8

Dataset overview | Output and supplemental files | Input files | Learn about Visium analysis

Batch download

Input files	File type	Size	md5sum
FASTQs	TAR	6.26 GB	1448d8c7b3e9c54992a826da45f5eb44
Image (TIFF)	TIF	21.6 MB	a0bc338a4c6389904f977d98e227e809
Probe Set	CSV	6.13 MB	a74f1583689de3c9f95164b2fc090194
Microscope image without fiducial frame (TIFF)	TIF	2.38 GB	21c055fb9290ae4b0c5c663ae8487a81

In Rstudio

```
##spatial transcriptomic- 10x visium##
#Najneen Rejwana
#Source :
https://www.10xgenomics.com/datasets/fresh-frozen-visium-on-cytassist-human-breast-cancer-probe-based-whole-transcriptome-profiling-2-standard


#-------------folder arrangement should be like this-------------------------
# G:/Rstudio/TRANSCRIPTOMICS/spatial/10x_visium/breast_cancer/
#  ├── filtered_feature_bc_matrix.h5
#  ├── raw_feature_bc_matrix.h5
# ├── spatial/
#    ├── tissue_lowres_image.png
# ├    ├── scalefactors_json.json   (Required for correct scaling)
# ├    ├── tissue_positions_list.csv (Required for spot positioning)



getwd()
setwd("G:/Rstudio/TRANSCRIPTOMICS/spatial/10x_visium/breast_cancer")

library(Seurat)
library(SeuratDisk)
library(ggplot2)
library(SeuratData)
library(patchwork)
library(tiff)
library(png)

#install.packages("Seurat")
#ls("package:Seurat")  # Lists all functions in Seurat

#install.packages("remotes")  # Install remotes if not already installed
#remotes::install_github("satijalab/seurat-data", force = TRUE)

#install.packages("tiff")
```

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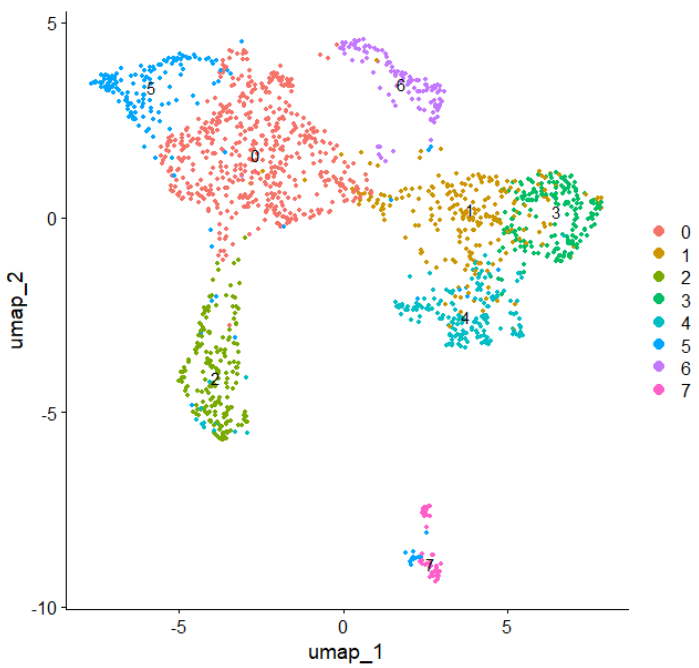
```
setwd("G:/Rstudio/TRANSCRIPTOMICS/spatial/10x_visium/breast_cancer")
list.files()
#[1] "CytAssist_Fresh_Frozen_Human_Breast_Cancer_analysis.tar.gz"
#[2] "CytAssist_Fresh_Frozen_Human_Breast_Cancer_cloupe.cloupe"
#[3] "CytAssist_Fresh_Frozen_Human_Breast_Cancer_filtered_feature_bc_matrix.h5"
#[4] "CytAssist_Fresh_Frozen_Human_Breast_Cancer_image.png"
#[5] "CytAssist_Fresh_Frozen_Human_Breast_Cancer_image.tif"
#[6] "CytAssist_Fresh_Frozen_Human_Breast_Cancer_metrics_summary.csv"
#[7] "CytAssist_Fresh_Frozen_Human_Breast_Cancer_raw_feature_bc_matrix.h5"
#[8] "CytAssist_Fresh_Frozen_Human_Breast_Cancer_spatial.tar.gz"
#[9] "spatial"
#[10] "spatial_10x_breast.R"
#[11] "tissue_lowres_image.png"

list.files("spatial")
#[1] "aligned_fiducials.jpg"     "aligned_tissue_image.jpg"  "cytassist_image.tiff"
#[4] "detected_tissue_image.jpg" "scalefactors_json.json"    "spatial_enrichment.csv"
#[7] "tissue_hires_image.png"    "tissue_lowres_image.png"   "tissue_positions.csv"



##Loading Spatial data into Seurat-----------------------------------------------------------------
data_dir <- getwd()
h5_file <- "CytAssist_Fresh_Frozen_Human_Breast_Cancer_filtered_feature_bc_matrix.h5"

# Load the spatial dataset
seurat_obj <- Load10X_Spatial(
  data.dir = data_dir,
  filename = h5_file,
  assay = "Spatial",
  slice = "breast_tissue",
  filter.matrix = TRUE
)

# Print summary of Seurat object
seurat_obj

#An object of class Seurat
#18085 features across 1657 samples within 1 assay
```

#Active assay: Spatial (18085 features, 0 variable features)
#1 layer present: counts
#1 spatial field of view present: breast_tissue

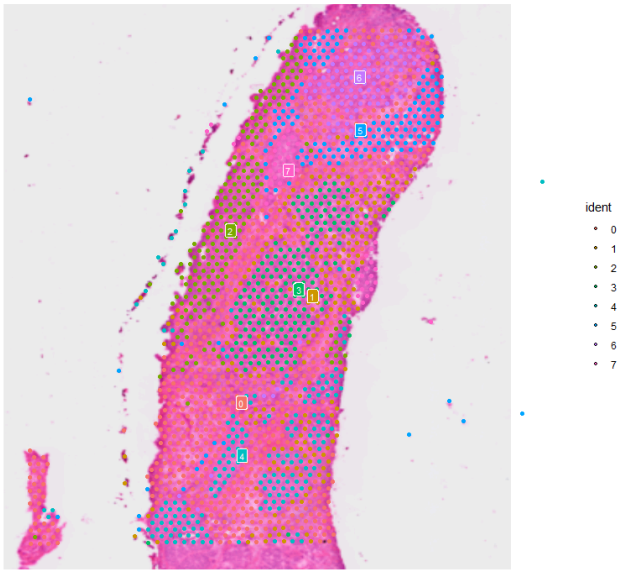
##Preprocessing data----------

```
# Normalize using SCTransform
seurat_obj <- SCTransform(seurat_obj, assay = "Spatial", verbose = FALSE)

# Dimensionality reduction
seurat_obj <- RunPCA(seurat_obj, verbose = FALSE)
seurat_obj <- FindNeighbors(seurat_obj, dims = 1:30)
seurat_obj <- FindClusters(seurat_obj, resolution = 0.5)
seurat_obj <- RunUMAP(seurat_obj, dims = 1:30)

# Plot UMAP
DimPlot(seurat_obj, reduction = "umap", label = TRUE)
```

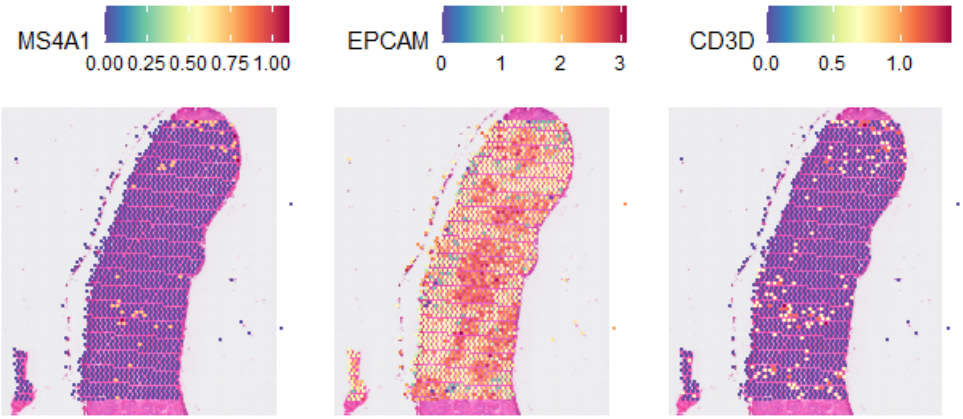


##Visualize Spatial data----------

# View tissue image with spots
SpatialDimPlot(seurat_obj, label = TRUE, label.size = 3)



# Plot marker gene expression in spatial context
SpatialFeaturePlot(seurat_obj, features = c("MS4A1", "EPCAM", "CD3D"))

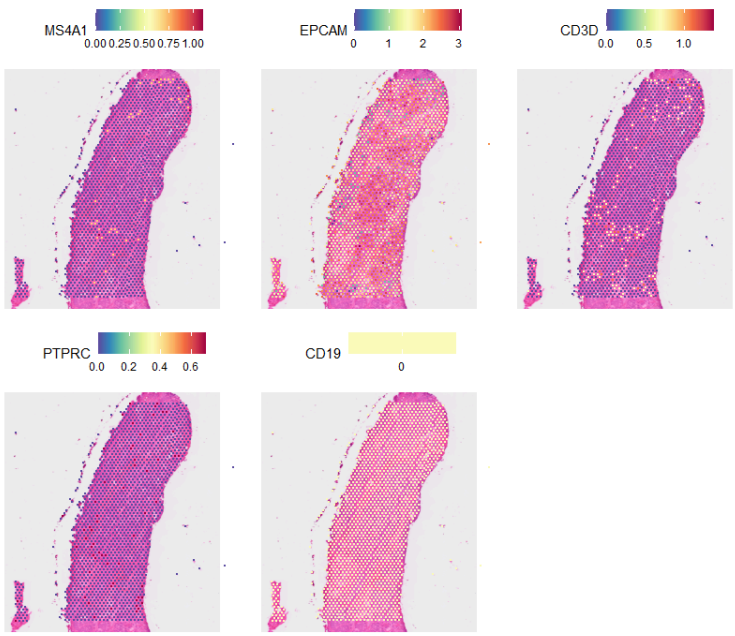


##Differential gene expression----------

# Find differentially expressed genes
markers <- FindAllMarkers(seurat_obj, min.pct = 0.25, logfc.threshold = 0.25)

View top marker genes

```
head(markers)
#              p_val          avg_log2F pct.1 pct.2 p_val_adj        cluster gene
#VIM    1.499643e-207   2.054761      1 0.917 2.251414e-203       0    VIM
#COL3A1 1.038675e-206   1.976601      1 0.979 1.559363e-202       0 COL3A1
#CD74   9.252261e-200   2.658123      1 0.923 1.389042e-195       0   CD74
#COL1A1 3.248479e-198   2.028522      1 1.000 4.876942e-194       0 COL1A1
#COL1A2 1.278452e-192   1.918285      1 0.949 1.919341e-188       0 COL1A2
#TIMP2  1.082596e-189   2.029384      1 0.795 1.625302e-185       0  TIMP2


# Define some marker genes for cell-type identification
marker_genes <- c("MS4A1", "EPCAM", "CD3D", "PTPRC", "CD19")

# Plot expression of these marker genes
SpatialFeaturePlot(seurat_obj, features = marker_genes)
```




```
# Visualizing expression of a gene in spatial context
SpatialFeaturePlot(seurat_obj, features = "MS4A1")
```

```
# Create a volcano plot—------------------
library(ggplot2)

# Add a column for significance based on p-value threshold
de_genes$significance <- ifelse(de_genes$p_val_adj < 0.05, "Significant", "Not Significant")

# Plot
ggplot(de_genes, aes(x = avg_log2FC, y = -log10(p_val_adj), color = significance)) +
  geom_point(alpha = 0.6) +
  scale_color_manual(values = c("Not Significant" = "red", "Significant" = "blue")) +
  theme_minimal() +
  ggtitle("Volcano Plot: Cluster 0 vs Cluster 1") +
  xlab("Log2 Fold Change") +
  ylab("-log10 Adjusted P-value") +
  theme(legend.position = "top")
```

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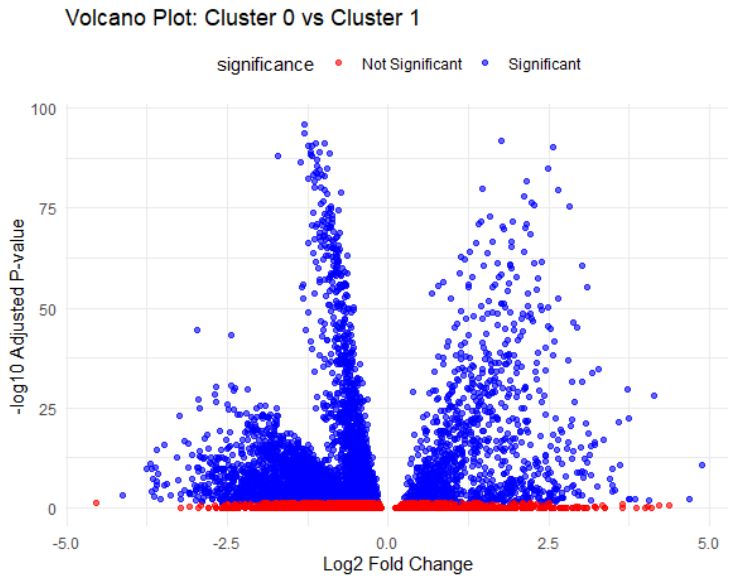


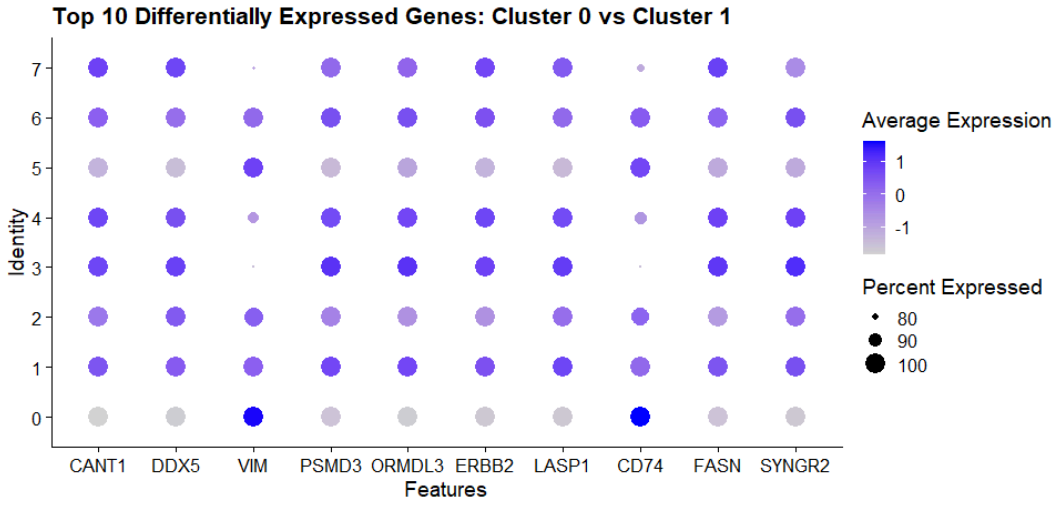
#Dot plot—----------

Get the top 10 genes by p-value

```
top_genes <- head(de_genes[order(de_genes$p_val_adj), ], 10)
```

Create a dot plot for the top 10 genes

```
top_gene_names <- rownames(top_genes)
DotPlot(seurat_obj, features = top_gene_names) +
  ggtitle("Top 10 Differentially Expressed Genes: Cluster 0 vs Cluster 1")
```

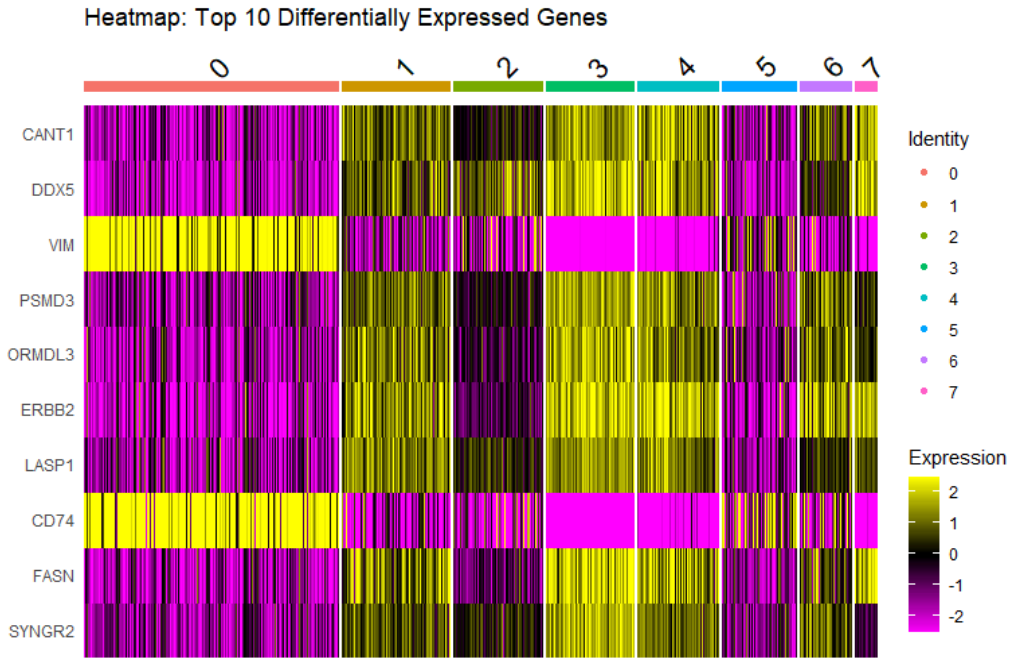


#Heatmap----------

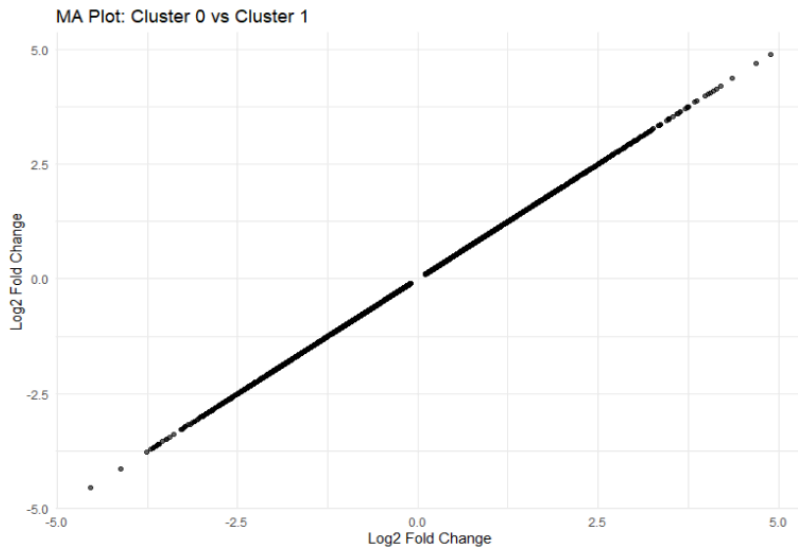
Select the top 10 genes based on p-value

```
top_genes_for_heatmap <- rownames(top_genes)
```

```
# Plot the heatmap
DoHeatmap(seurat_obj, features = top_genes_for_heatmap) +
  ggtitle("Heatmap: Top 10 Differentially Expressed Genes")
```




```
#MA Plot-----------------------------------------------------
ggplot(de_genes, aes(x = avg_log2FC, y = avg_log2FC)) +
  geom_point(alpha = 0.6) +
  ggtitle("MA Plot: Cluster 0 vs Cluster 1") +
  xlab("Log2 Fold Change") +
  ylab("Log2 Fold Change") +
  theme_minimal()
```




```
##Number of genes-----------------------------------------------------------------
#Total number of genes in de_genes
nrow(de_genes)
#[1] 11300


#Number of significant DEGs (Adjusted p-value < 0.05)
sum(de_genes$p_val_adj < 0.05, na.rm = TRUE)
#[1] 4701


#Number of upregulated and downregulated genes
# Count upregulated genes (log2FC > 0 & adj p-value < 0.05)
sum(de_genes$avg_log2FC > 0 & de_genes$p_val_adj < 0.05, na.rm = TRUE)
#[1] 1105

# Count downregulated genes (log2FC < 0 & adj p-value < 0.05)
sum(de_genes$avg_log2FC < 0 & de_genes$p_val_adj < 0.05, na.rm = TRUE)
#[1] 3596



#Filter Upregulated and Downregulated Genes-------------------
```

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```
# Define thresholds
logFC_threshold <- 0
pval_threshold <- 0.05  # Adjusted p-value threshold

# Filter upregulated genes (log2FC > 0 & adj p-value < 0.05)
upregulated_genes <- de_genes[de_genes$avg_log2FC > logFC_threshold &
de_genes$p_val_adj < pval_threshold, ]

# Filter downregulated genes (log2FC < 0 & adj p-value < 0.05)
downregulated_genes <- de_genes[de_genes$avg_log2FC < logFC_threshold &
de_genes$p_val_adj < pval_threshold, ]


# Save upregulated genes to CSV
write.csv(upregulated_genes, "upregulated_genes_breast_spatial.csv", row.names = TRUE)

# Save downregulated genes to CSV
write.csv(downregulated_genes, "downregulated_genes_breast_spatial.csv", row.names =
TRUE)
```

upregulated_genes.csv

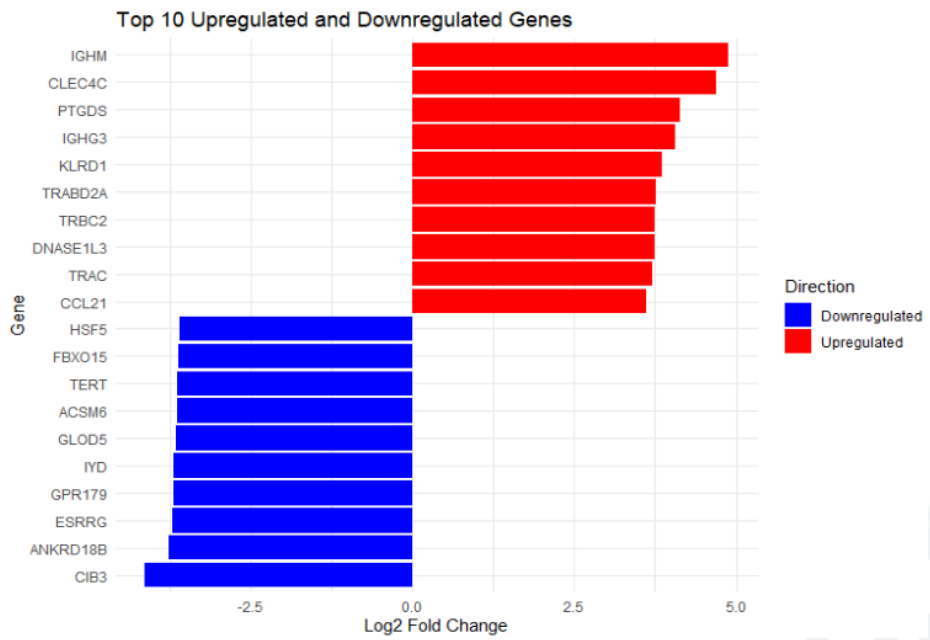
	A	B	C	D	E	F	G
1		p_val	avg_log2FC	pct.1	pct.2	p_val_adj	significance
2	VIM	1.04164683746839e-96	1.75656256909682	1	1	1.5638243970913e-92	Significant
3	CD74	2.95425431168261e-95	2.5719065751083	1	1	4.43522199812911e-91	Significant
4	ZFP36L2	8.5710553996594e-90	2.4929806902014	1	0.85	1.28677254715087e-85	Significant
5	FTL	9.19140347249744e-87	2.16019609862739	1	1	1.37990540332604e-82	Significant
6	COL3A1	1.00366482794148e-84	1.46441163949769	1	1	1.50680200618854e-80	Significant
7	CEBPD	2.07119985283312e-84	2.64177153190465	0.994	0.742	3.10949233905836e-80	Significant
8	KCTD12	5.05144493608696e-83	2.10819490354634	1	0.88	7.58373428254735e-79	Significant
9	MAFB	3.15911422443392e-81	2.23653727670376	1	0.893	4.74277818514264e-77	Significant
10	C1R	1.05306277844219e-80	2.26413860259642	0.996	0.807	1.58096314927526e-76	Significant
11	CXCR4	1.69072374082625e-80	2.8251540227109	0.994	0.734	2.53828355210244e-76	Significant
12	TIMP2	9.2172818533933e-78	1.58691219673777	1	0.953	1.38379052464994e-73	Significant
13	LY6E	1.21632162969097e-76	1.92597923580125	1	0.91	1.82606366265506e-72	Significant
14	COL6A3	1.44041220040356e-76	1.4480401388633	1	1	2.16249083646586e-72	Significant
15	C1QC	7.19376271886919e-76	2.14787086092359	1	0.871	1.07999959698383e-71	Significant
16	COL1A1	8.20552573934262e-76	1.40889451397981	1	1	1.23189557924751e-71	Significant
17	IGFBP4	2.67112664757336e-75	1.77641838136937	0.998	0.893	4.01016243600189e-71	Significant
18	CCN1	4.28357759842335e-75	2.13050103780844	0.998	0.854	6.43093504851298e-71	Significant
19	TXNIP	1.2855741166403e-74	1.80097335252238	0.998	0.953	1.93003242131209e-70	Significant
20	C1QA	2.3597683976689e-73	2.21749314425012	0.998	0.88	3.54272029542033e-69	Significant

Downregulated_genes.csv

	A	B	C	D	E	F	G
1		p_val	avg_log2FC	pct.1	pct.2	p_val_adj	significance
2	CANT1	4.73146475474292e-101	-1.294442999	1	1	7.10334803629555e-97	Significant
3	DDX5	8.71882587476787e-99	-1.298846865	1	1	1.3089573285789e-94	Significant
4	PSMD3	3.75690676117549e-96	-0.989080887	1	1	5.64024412055276e-92	Significant
5	ORMDL3	4.04939424396924e-96	-1.120214186	1	1	6.07935557847102e-92	Significant
6	ERBB2	1.17124059537347e-95	-1.248493355	1	1	1.75838350583419e-91	Significant
7	LASP1	1.35315789868835e-95	-1.187889806	1	1	2.03149595330083e-91	Significant
8	FASN	5.38923504868901e-94	-1.199792841	1	1	8.09085857859682e-90	Significant
9	SYNGR2	6.14080475811146e-94	-1.074774565	1	1	9.21919018335274e-90	Significant
10	MT-CO2	9.26485634251226e-94	-0.918233964	1	1	1.39093288270137e-89	Significant
11	ARHGAP8	3.53564506161648e-93	-1.204988897	0.998	1	5.30806393100483e-89	Significant
12	AMBP	3.61676973718468e-93	-1.716423432	0.993	1	5.42985640643536e-89	Significant
13	RHOBTB3	4.54134700145002e-93	-1.17757814	1	1	6.81792425327691e-89	Significant
14	SGSH	4.0070398547132e-92	-1.105993436	1	1	6.01576893388093e-88	Significant
15	ACSF2	2.52332699569336e-91	-1.356181285	0.998	1	3.78827081863444e-87	Significant
16	AKAP1	2.2898002411637e-90	-1.115220545	1	1	3.43767710205906e-86	Significant
17	MLLT6	9.81972212093899e-90	-0.945049662	1	1	1.47423488201657e-85	Significant
18	CYB561	1.54420724850544e-89	-1.071695067	1	1	2.31831834218122e-85	Significant
19	GRB7	7.64426492609744e-89	-1.134921004	1	1	1.14763349335501e-84	Significant
20	MLPH	1.32091787285669e-88	-1.098831874	1	1	1.98309400251975e-84	Significant

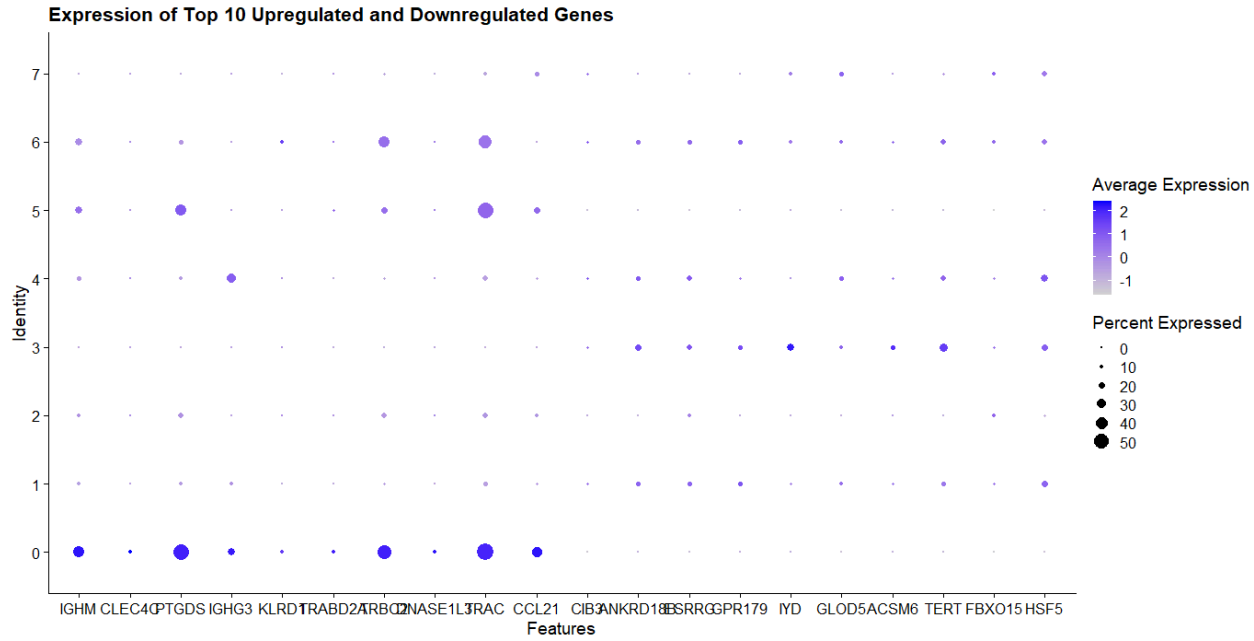
```
#Extract Top 10 Upregulated and Downregulated Genes----------------------
# Define thresholds
logFC_threshold <- 0
pval_threshold <- 0.05  # Adjusted p-value threshold

# Filter significant upregulated and downregulated genes
upregulated_genes <- de_genes[de_genes$avg_log2FC > logFC_threshold &
de_genes$p_val_adj < pval_threshold, ]
downregulated_genes <- de_genes[de_genes$avg_log2FC < logFC_threshold &
de_genes$p_val_adj < pval_threshold, ]

# Select top 10 upregulated genes (sorted by log2 fold change)
top_10_up <- upregulated_genes[order(-upregulated_genes$avg_log2FC), ][1:10, ]

# Select top 10 downregulated genes (sorted by log2 fold change)
top_10_down <- downregulated_genes[order(downregulated_genes$avg_log2FC), ][1:10, ]

# Print the gene names
print("Top 10 Upregulated Genes:")
print(rownames(top_10_up))
# [1] "IGHM"    "CLEC4C"   "PTGDS"    "IGHG3"    "KLRD1"    "TRABD2A"  "TRBC2"
"DNASE1L3" "TRAC"
#[10] "CCL21"
```

```
print("Top 10 Downregulated Genes:")
print(rownames(top_10_down))
# [1] "CIB3"     "ANKRD18B" "ESRRG"    "GPR179"   "IYD"      "GLOD5"    "ACSM6"
"TERT"     "FBXO15"
#[10] "HSF5"


# Save top upregulated genes
write.csv(top_10_up, "top_10_upregulated_genes.csv", row.names = TRUE)

# Save top downregulated genes
write.csv(top_10_down, "top_10_downregulated_genes.csv", row.names = TRUE)




#Visualize using Bar plot-------------------------------------------------------
library(ggplot2)

# Combine top 10 upregulated and downregulated genes into one dataframe
top_genes_combined <- rbind(
  data.frame(Gene = rownames(top_10_up), Log2FC = top_10_up$avg_log2FC, Direction =
"Upregulated"),
  data.frame(Gene = rownames(top_10_down), Log2FC = top_10_down$avg_log2FC, Direction
= "Downregulated")
)

# Plot
ggplot(top_genes_combined, aes(x = reorder(Gene, Log2FC), y = Log2FC, fill = Direction)) +
  geom_bar(stat = "identity") +
  coord_flip() +
  theme_minimal() +
  labs(title = "Top 10 Upregulated and Downregulated Genes",
       x = "Gene",
       y = "Log2 Fold Change") +
  scale_fill_manual(values = c("Upregulated" = "red", "Downregulated" = "blue"))
```



#Dot plot----------

```
DotPlot(seurat_obj, features = c(rownames(top_10_up), rownames(top_10_down))) +
  ggtitle("Expression of Top 10 Upregulated and Downregulated Genes")
```

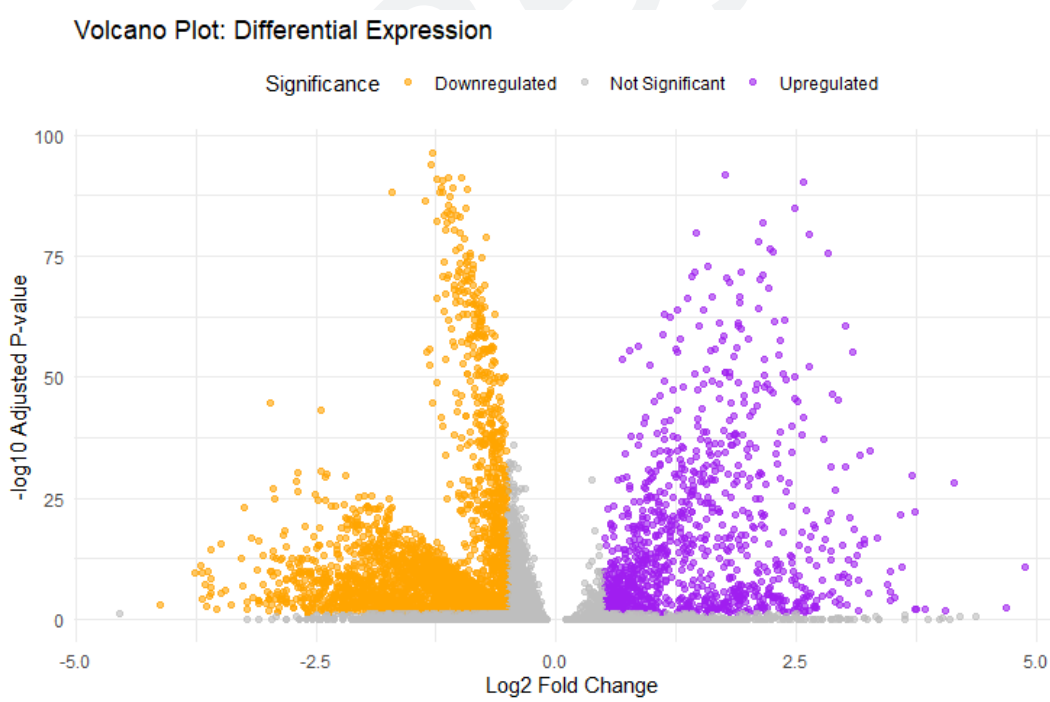


##volcano plot for upregulated & downregulated genes---------

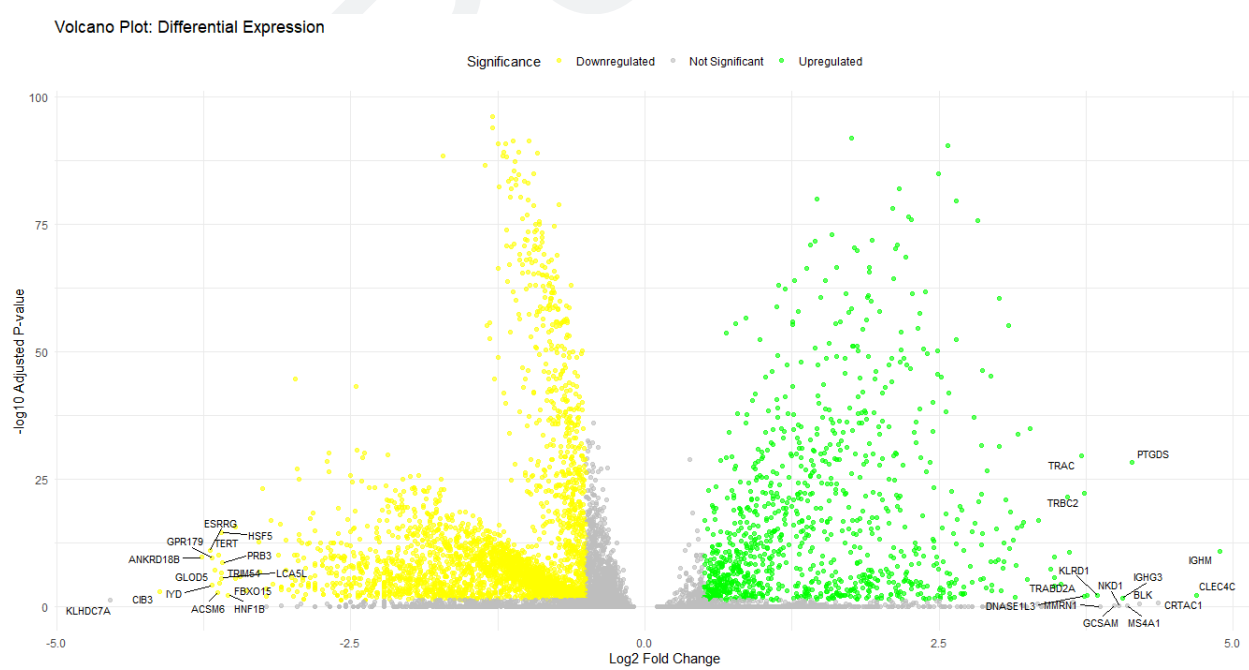
```
library(ggplot2)
```

Define significance thresholds

```
logFC_threshold <- 0.5  # Change if needed
pval_threshold <- 0.05

# Add a new column to classify genes as Upregulated, Downregulated, or Not Significant
de_genes$Significance <- ifelse(
  de_genes$p_val_adj < pval_threshold & de_genes$avg_log2FC > logFC_threshold,
"Upregulated",
  ifelse(de_genes$p_val_adj < pval_threshold & de_genes$avg_log2FC < -logFC_threshold,
"Downregulated", "Not Significant")
)

# Create the volcano plot
ggplot(de_genes, aes(x = avg_log2FC, y = -log10(p_val_adj), color = Significance)) +
  geom_point(alpha = 0.6, size = 1.5) +
  scale_color_manual(values = c("Upregulated" = "purple", "Downregulated" = "orange", "Not
Significant" = "gray")) +
  theme_minimal() +
  labs(title = "Volcano Plot: Differential Expression",
       x = "Log2 Fold Change",
       y = "-log10 Adjusted P-value") +
  theme(legend.position = "top")
```



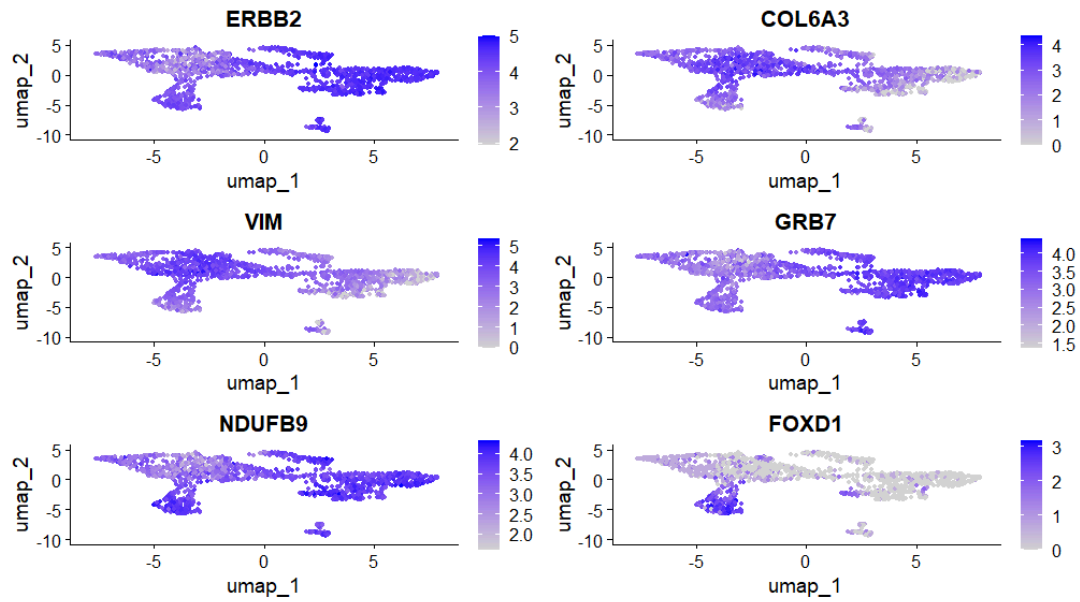
```
# Select top 15 most significant genes from both upregulated and downregulated categories
top_up <- rownames(head(de_genes[order(-de_genes$avg_log2FC, de_genes$p_val_adj), ], 15))
top_down <- rownames(head(de_genes[order(de_genes$avg_log2FC, de_genes$p_val_adj), ],
15))
top_genes <- c(top_up, top_down)

# Create the volcano plot with black text labels
ggplot(de_genes, aes(x = avg_log2FC, y = -log10(p_val_adj), color = Significance)) +
  geom_point(alpha = 0.6, size = 1.5) +
  scale_color_manual(values = c("Upregulated" = "green", "Downregulated" = "yellow", "Not
Significant" = "gray")) +
  geom_text_repel(data = subset(de_genes, rownames(de_genes) %in% top_genes),
                  aes(label = rownames(subset(de_genes, rownames(de_genes) %in% top_genes))),
                  color = "black",  # Set text color to black
                  size = 3, max.overlaps = 15) +
  theme_minimal() +
  labs(title = "Volcano Plot: Differential Expression",
       x = "Log2 Fold Change",
       y = "-log10 Adjusted P-value") +
  theme(legend.position = "top")
```



#Feature plot (for specific genes)----------

library(Seurat)

# Visualize expression of specific genes in Seurat object
FeaturePlot(seurat_obj, features = c("ERBB2", "COL6A3", "VIM", "GRB7", "NDUFB9", "FOXD1"))



##Spatial Gene Expression Enrichment (GESA)----------
#Enrichment of Specific Pathways: Use gene set enrichment analysis (GSEA) to identify if certain biological pathways are enriched in different spatial regions of the tissue.

install.packages("clusterProfiler")
library(clusterProfiler)

#Suppose for pyrimidine pathway----------
#To perform a Gene Set Enrichment Analysis (GSEA) for the pyrimidine pathway, you need to first identify the genes associated with this pathway. One common resource for this is the KEGG pathway database, which provides a collection of gene sets for various biological pathways, including the pyrimidine pathway.

Check the available identities in this Seurat object

```
levels(seurat_obj)
#[1] "0" "1" "2" "3" "4" "5" "6" "7"
```

```
# Perform differential expression between Cluster 0 and Cluster 1
de_genes <- FindMarkers(seurat_obj, ident.1 = "0", ident.2 = "1")
print(de_genes)
```

```
                p_val avg_log2FC pct.1 pct.2    p_val_adj
CANT1    4.731465e-101 -1.2944430 1.000 1.000 7.103348e-97
DDX5      8.718826e-99 -1.2988469 1.000 1.000 1.308957e-94
VIM       1.041647e-96  1.7565626 1.000 1.000 1.563824e-92
PSMD3     3.756907e-96 -0.9890809 1.000 1.000 5.640244e-92
ORMDL3    4.049394e-96 -1.1202142 1.000 1.000 6.079356e-92
ERBB2     1.171241e-95 -1.2484934 1.000 1.000 1.758384e-91
LASP1     1.353158e-95 -1.1878898 1.000 1.000 2.031496e-91
CD74      2.954254e-95  2.5719066 1.000 1.000 4.435222e-91
FASN      5.389235e-94 -1.1997928 1.000 1.000 8.090859e-90
SYNGR2    6.140805e-94 -1.0747746 1.000 1.000 9.219190e-90
MT-CO2    9.264856e-94 -0.9182340 1.000 1.000 1.390933e-89
ARHGAP8   3.535645e-93 -1.2049889 0.998 1.000 5.308064e-89
AMBP      3.616770e-93 -1.7164234 0.993 1.000 5.429856e-89
RHOBTB3   4.541347e-93 -1.1775781 1.000 1.000 6.817924e-89
SGSH      4.007040e-92 -1.1059934 1.000 1.000 6.015769e-88
ACSF2     2.523327e-91 -1.3561813 0.998 1.000 3.788271e-87
AKAP1     2.289800e-90 -1.1152205 1.000 1.000 3.437677e-86
ZFP36L2   8.571055e-90  2.4929807 1.000 0.850 1.286773e-85
MLLT6     9.819722e-90 -0.9450497 1.000 1.000 1.474235e-85
CYB561    1.544207e-89 -1.0716951 1.000 1.000 2.318318e-85
GRB7      7.644265e-89 -1.1349210 1.000 1.000 1.147633e-84
MLPH      1.320918e-88 -1.0988319 1.000 1.000 1.983094e-84
TOB1      2.397173e-88 -1.1607018 1.000 1.000 3.598875e-84
SPINT2    3.068166e-88 -1.0240548 1.000 1.000 4.606237e-84
CASC3     6.433002e-88 -0.9934367 1.000 1.000 9.657866e-84
BCAM      1.619758e-87 -1.0940068 1.000 1.000 2.431743e-83
RGS16     3.577011e-87 -1.2418901 1.000 1.000 5.370166e-83
FTL       9.191403e-87  2.1601961 1.000 1.000 1.379905e-82
FXYD3     9.418388e-87 -1.1383126 0.998 1.000 1.413983e-82
CCDC47    3.381643e-85 -1.0581812 1.000 1.000 5.076860e-81
GSDMB     3.834943e-85 -1.1477928 1.000 1.000 5.757400e-81
COL3A1    1.003665e-84  1.4644116 1.000 1.000 1.506802e-80
CDK12     1.463680e-84 -0.9941364 1.000 1.000 2.197422e-80
CEBPD     2.071200e-84  2.6417715 0.994 0.742 3.109492e-80
MT-ND4    1.060708e-83 -0.7319855 1.000 1.000 1.592440e-79
PGAP3     1.522146e-83 -0.9488019 1.000 1.000 2.285197e-79
KCTD12    5.051445e-83  2.1081949 1.000 0.880 7.583734e-79
PCGF2     9.916619e-82 -1.0003660 1.000 1.000 1.488782e-77
MAFB      3.159114e-81  2.2365373 1.000 0.893 4.742778e-77
```

```
DCXR       6.141684e-81 -1.0401417 1.000 1.000 9.220509e-77
C1R        1.053063e-80  2.2641386 0.996 0.807 1.580963e-76
CXCR4      1.690724e-80  2.8251540 0.994 0.734 2.538284e-76
ESR1       2.413985e-80 -0.8948692 1.000 1.000 3.624116e-76
RPTOR      7.005905e-80 -0.8966002 1.000 1.000 1.051796e-75
BAG1       7.922718e-80 -0.9337853 1.000 1.000 1.189438e-75
MT-ATP6    1.835639e-79 -0.7749585 1.000 1.000 2.755844e-75
CCDC78     1.065376e-78 -1.1735919 1.000 1.000 1.599449e-74
MIEN1      1.097002e-78 -0.9194805 1.000 1.000 1.646929e-74
PBX1       2.421865e-78 -0.9955469 0.998 1.000 3.635946e-74
KRT8       2.693959e-78 -0.9419501 1.000 1.000 4.044440e-74
H2AFJ      3.579931e-78 -0.8717772 1.000 1.000 5.374551e-74
TIMP2      9.217282e-78  1.5869122 1.000 0.953 1.383791e-73
PSMC5      3.356267e-77 -0.8711524 1.000 1.000 5.038764e-73
GPRC5A     7.332544e-77 -1.0213274 1.000 1.000 1.100835e-72
EIF4A3     7.975654e-77 -0.9287229 1.000 1.000 1.197385e-72
LY6E       1.216322e-76  1.9259792 1.000 0.910 1.826064e-72
SREBF1     1.232528e-76 -1.0055012 1.000 1.000 1.850394e-72
COL6A3     1.440412e-76  1.4480401 1.000 1.000 2.162491e-72
AFF3       2.676233e-76 -0.8765981 1.000 1.000 4.017829e-72
TFF3       6.090292e-76 -1.1165724 0.998 1.000 9.143355e-72
C1QC       7.193763e-76  2.1478709 1.000 0.871 1.080000e-71
COL1A1     8.205526e-76  1.4088945 1.000 1.000 1.231896e-71
—----------------------------------------------------
—----------------------------------------------------
```

```
#Prepare a ranked gene list-----
# Create a ranked gene list based on avg_log2FC
ranked_genes <- de_genes[, "avg_log2FC"]
names(ranked_genes) <- rownames(de_genes)

# Sort genes from highest to lowest log fold change
ranked_genes <- sort(ranked_genes, decreasing = TRUE)


#Obtain pyrimidine pathway from KEGG---------------
if (!requireNamespace("clusterProfiler", quietly = TRUE)) install.packages("clusterProfiler")
if (!requireNamespace("org.Hs.eg.db", quietly = TRUE)) install.packages("org.Hs.eg.db")

library(clusterProfiler)
library(org.Hs.eg.db)  # Human gene annotation
```

```
#Retrieve KEGG Pathway Gene Sets-----
# Download KEGG pathways for humans
kegg_pathways <- enrichKEGG(gene = NULL, organism = "hsa", keyType = "kegg")

# Get ranked gene list (logFC and p-value)
ranked_genes <- de_genes[, "avg_log2FC"]
names(ranked_genes) <- rownames(de_genes)
ranked_genes <- sort(ranked_genes, decreasing = TRUE)

# Extract Entrez IDs from the ranked genes (make sure your data contains these)
gene_list <- bitr(names(ranked_genes), fromType = "SYMBOL", toType = "ENTREZID",
OrgDb = org.Hs.eg.db)

# Filter Entrez IDs and use for KEGG enrichment
entrez_gene_list <- gene_list$ENTREZID



# Run KEGG enrichment
kegg_pathways <- enrichKEGG(gene = entrez_gene_list, organism = "hsa")

# Check the results of KEGG enrichment
head(kegg_pathways@result)



#Extract Pyrimidine Pathway Genes----
# Extract the gene set for Pyrimidine metabolism (KEGG ID: hsa00240)
pyrimidine_genes <- kegg_pathways@result[kegg_pathways@result$ID == "hsa00240",
"geneID"]

# Split and convert the gene IDs to gene symbols
pyrimidine_gene_list <- bitr(strsplit(pyrimidine_genes, "/")[[1]], fromType = "ENTREZID",
toType = "SYMBOL", OrgDb = org.Hs.eg.db)

# Check extracted genes
```

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```
head(pyrimidine_gene_list)
# ENTREZID SYMBOL
#1    51733   UPB1
#2   129607  CMPK2
#3     1806   DPYD
#4     1890   TYMP
#5     4907   NT5E
#6      953 ENTPD1
```

```
#Run GESA---
gsea_results <- GSEA(
  geneList = ranked_genes,  # Ranked gene list from FindMarkers
  TERM2GENE = pyrimidine_gene_list,
  pvalueCutoff = 0.05,
  minGSSize = 10,
  maxGSSize = 500,
  verbose = TRUE
)

# Check GSEA results
head(gsea_results@result)
#[1] ID          Description     setSize         enrichmentScore NES             pvalue
#[7] p.adjust        qvalue
#<0 rows> (or 0-length row.names)
```

##That means no significant pathways were identified by GSEA based on the pyrimidine pathway gene set.

```
#Save processed data-----------------------------------------------------------
saveRDS(seurat_obj, file = "spatial_breast_cancer_seurat.rds")
```

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